



Response to the National Science Foundation's Request for Information on the Development of an Artificial Intelligence Action Plan

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Queer in AI submits this document in response to the National Science Foundation's (NSF) Request for Information (RFI) on the development of an artificial intelligence (AI) action plan.

Queer in AI is a part of oSTEM, a 501(c)(3) non-profit professional association for queer people in Science, Technology, Engineering, Mathematics (STEM) fields. Queer in AI was established by queer scientists working in artificial intelligence and machine learning (AI/ML) with the mission to make the AI community a safe and inclusive place that welcomes, supports, and values queer people. A crucial part of our mission is to raise awareness of queer issues in the general AI community and to encourage and highlight research on these problems. We are made up of scientists and experts working directly on AI across industry, academia, and civil society.

We have deep collective knowledge on how to incorporate safe, privacy-preserving, secure, assurable processes throughout the AI lifecycle. We maintain a belief that safe, inclusive frameworks must be developed in tandem with AI systems. AI innovation shouldn't be at the expense of harm (psychological, physical, and institutional, or individual). We welcome the opportunity to submit these comments.

We present a set of action items we believe essential to the development of an AI action plan. They are summarized as follows:

1. **Increase broad participation in AI development from non-technical and marginalized communities.** Increased participation encourages robust AI development practices.
2. **Encourage explicit consent practices when gathering and using personal data in AI development.** Privacy and security concerns towards AI systems are multifold and affect communities differently.

3. **Avoid using AI towards tasks that are pseudo-scientific or ambiguously defined.** AI systems may encourage legitimacy towards ill-defined end tasks resulting in harm or misuse.
4. **Limit the use of AI in law enforcement, weapons development, and immigration.** Bias and uncertainty caused at different levels (i.e., dataset curation, task description, model architecture and behavior) are present in AI systems designed for narrow and large end tasks. These areas are considered high-risk and insufficient regulation or research knowledge exists to prevent model harm or misuse.

1. Encourage broad participation especially from non-technical communities.

We continue to ask for increased engagement with queer people. Feedback from queer people at all stages of the AI lifecycle can highlight concerns in AI systems and aids in more robust design¹. Queer experiences vary by individual and community; this diversity is essential for robust AI systems. Queer people are affected throughout the AI lifecycle, be it erasure through binary language in collected data, treatment as outliers during training, or surveillance through deployed systems. Queer people represent over 7% of the US population² and have faced long, painful histories of exclusion from and targeting by science and technology^{3,4}.

We encourage involving impacted communities in AI governance and development. We suggest first steps for incorporating broad participation⁵:

1. Government and private sector use of AI systems may benefit from due process. Due process by individuals or democratic mechanisms increases safety and accountability over decisions made by AI systems. Due process rights encourage autonomy by giving marginalized groups pathways to challenge harm from AI systems (i.e., AI decisions, privacy-infringing data curation). We encourage procedural frameworks for addressing safety and security of marginalized groups. These frameworks should offer mechanisms for feedback and pathways to challenge potential harms caused by AI decisions.
2. Risk management frameworks (RMF) such as the National Institute of Standards and Technology AI Risk Management Framework (NIST RMF) still serve as a strong framework to minimize harm across different risk levels. We believe that risk should be assessed by external parties and the public through mechanisms such as requests for comment (RFC). RMFs can work in tandem with rights-based data governance approaches to encourage participation from non-technical communities. Public-private participation in AI governance may also benefit from feedback at different levels (e.g., agency level).

We believe it's beneficial to:

- Require AI developers to engage with queer communities when designing and developing AI tools and consult them about how they are impacted and represented. It is important to also financially and psychologically support marginalized communities when involving them in participatory design for AI⁶.
- Provide developers, evaluators, users, and the general public with AI education materials, with an emphasis on the specific harms faced by queer people (e.g., outing, misgendering, erasure).
- Consider targeted outreach to communities not heard from during RFI and Request for Comment (RFC) periods.

2. Privacy Needs vary. Privacy considerations should be made throughout the AI lifecycle

Privacy needs vary by marginalized community and individual. Doxxing or outing of queer individuals and anti-queer legislation may necessitate those queer perspectives be protected (e.g., noised, anonymized). The harms included here are not exhaustive and are frequently evolving.

Respecting consent and privacy while diversifying AI data sources is critical. We need to improve training data curation practices to be more queer-inclusive. AI learns to reproduce and amplify patterns in large amounts of training data. Many AI technologies are trained on language, image, audio, and video data that are scraped from the internet; however, the internet is filled with homophobic, transphobic, racist, sexist, and abusive content, which manifests in training datasets, like LAION⁷. AI should only ever be trained on data that was obtained with affirmative and meaningful opt-in consent. In particular, queer data subjects should be informed of the specific harms that they may experience due to the inclusion of their writing, images, audio, or video in AI training datasets. For example, dataset search tools and AI memorization of training data can cause queer individuals to have their visibility heightened or be outed, threatening their privacy, safety, and employment. Furthermore, there should be clear, easy, and effective mechanisms for opting out of including one's data in a dataset at any time, and consent needs to be re-obtained every time the terms of data usage change. Moreover, contextual and effective privacy-preservation measures need to be employed to protect queer data. "Making AI more inclusive" is not sufficient justification for bypassing consensual and privacy-protecting data practices (e.g., via scraping posts/images from social media). We support meaningful consent practices by providing information that is in the interest of affected groups⁶.

Surveillance. We consider surveillance from the stance that it can exploit consent norms and invade individual privacy. Before 2023⁸, US Airport scanners and Transportation Security Association (TSA) officers operated heavily on binary cisgender

characteristics. Many accounts voiced concern from airport scanners flagging a high number of trans people and subjecting them to higher rates of privacy invasion⁹. We support safeguards within the US government (e.g., Fair Information Practice Principles), and would like similar safeguards for public-private partnerships. Public-private partnerships operate under ill-defined security regulation that blurs and expands the surveillance capacity of government agencies. Surveillance capabilities are dependent on the kinds and quantity of information collected across government agencies and the private sector alike.

AI can facilitate surveillance practices in cases where technology is shared (e.g., shared household computers or networks). AI targeted ads can unsafely disclose sexual orientation or gender identity to a larger audience.

We encourage frameworks that encourage safe AI development with respect to marginalized populations. **We highlight the following practices:**

1. Informed Consent encourages Autonomy
 - 1.1. End-users should perform consent within the context of their beliefs, needs, goals, and desires¹⁰. This is often at odds with simplicity in the information presented.
2. Encourage Transparent Development and Deployment Practices
 - 2.1. Cases of copyright infringement and harm by AI systems are hard to legislate due to the opacity of AI systems. Datasets used to train systems and development practices may be proprietary. Models may be inherently opaque in how they reach a decision.
 - 2.2. Opaque AI systems are often harder to trust or justify due to their inability to show cause-effect relationships
3. Encourage responsible, risk-aware Terms of Use Agreements like [Responsible AI Licenses](#)
4. Involve transparent use frameworks such as [Data and Society's Algorithmic Impact Assessment](#).

5. AI systems should avoid pseudo-scientific or ambiguous task descriptions

Government agencies should redlight the use of AI systems that make pseudo-scientific predictions. We specifically highlight the use of emotion detection¹¹ and gender recognition systems which are fundamentally flawed and further surveil and infringe on the civil rights of queer people. Furthermore, AI has given a dangerous veneer of legitimacy to physiognomy and phrenology, including using computer vision to identify queer people^{12,13} and infer gender from faces^{14,15,16}.

6. Limit the use of AI in Law Enforcement, Immigration, and Weapons Development

We are against the use of AI tools by law enforcement. Current AI applications in law enforcement range from notetaking to predictive policing¹⁷ and transcription of prisoner calls¹⁸. Furthermore, any AI systems that are used by law enforcement must be stringently audited. We encourage algorithmic auditing and mechanisms that work to limit aggressive surveillance to the queer community. AI systems in this area must be careful not to further historical social harms. There are long histories of data and surveillance being used by law enforcement¹⁷ to harm queer people, such as *Plaxico v. Michael* (1999)¹⁹ and the practice of “fairy shaking” by DC police where vehicle license data was used to extort queer people^{20,21}.

We support the following practices:

1. Internal Auditing²²
 - 1.1. Internal auditing should involve product developers, internal audit teams, management, and other stakeholders before deployment of a product (i.e., AI system).
 - 1.2. The scope of the audit should consider product requirements, discussion of core AI principles (e.g., robustness, security, privacy, fairness, etc.), an ethical review of the end use case(s), and review of the social impact of the AI system. Risk analysis should be performed with consideration of ethical implications.
2. External Auditing and Third-Party Oversight²³
 - 2.1. The scope of the audit should consider which threats to address (e.g., which demographic groups may be most affected by a product (i.e., AI system))
 - 2.2. External auditors should consider factors beyond benchmark performance. It is necessary to consider the whole of the AI development process in auditing (e.g., predatory data gathering practices, test design, documentation, guardrails, etc. must all bear weight in addition to final model performance).
 - 2.3. Minimize over-reliance on benchmarks. Benchmarks show a limited view of the system and must be used in tandem with previously mentioned artifacts (e.g., documentation, test design, etc.) to paint a clear picture.
 - 2.4. Consider how privacy, group representation, group fairness, and intersectionality are handled throughout the AI development lifecycle in addition to the final system.

Respectfully,
The Organizers of Queer in AI

Endnotes

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